



iJRASET

International Journal For Research in
Applied Science and Engineering Technology



INTERNATIONAL JOURNAL FOR RESEARCH

IN APPLIED SCIENCE & ENGINEERING TECHNOLOGY

Volume: 13 Issue: VII Month of publication: July 2025

DOI: <https://doi.org/10.22214/ijraset.2025.73384>

www.ijraset.com

Call:  08813907089

E-mail ID: ijraset@gmail.com

Smart Online Voting System using Face Authentication

Padma Priya A¹, Andruess K², Arun Prakash P³, Kavin P⁴

Computer Science and Engineering, SSM Institute of Engineering and Technology, Dindigul, India

Abstract: *The electoral process is the foundation of republic and governance. Within the last many decades, the election system has effectively experienced a number of variations. Despite having the largest maturity rule government in the world, India still uses either secret ballot voting (SBV) or electronic voting machines (EVM), both of which are precious, labour ferocious, and extravagant. The current system simply checked relating documents, adding the liability of fraudulent votes. The main thing of this system is to give an online voting system that, by using a camera for face recognition and OTP generation, will help to reduce namer fraud in homemade voting systems and earlier duplications of online voting. also, we're erecting a system of remote voting. to choosers who are unfit to travel to their birthplace's polling place. To guarantee the responsibility of the device, we're supplying software that has multiple layers of verification, including face recognition or verification, OTP verification, and confirmation data. Only after being authenticated and matched with the handed namer database may a single namer access the system. The namer will be suitable to continue choosing their chosen seeker from the panel once the corresponding face has been matched with the data handed.*

Keywords: Smart Voting System, Facial Recognition, OTP, Voter ID.

I. INTRODUCTION

The electoral process is a foundation of popular governance. still, traditional voting styles like paper ballots and EVMs face high costs, logistical challenges, and fraud pitfalls. As noted in the IRJET paper Smart Voting System Using Face Recognition and OTP[2], current systems calculate heavily on ID- grounded verification, which is vulnerable to phony . also, Smart Voting System Using Facial Discovery[5] highlights the challenges faced by remote voters. To address these issues, experimenters have proposed secure online voting systems using biometric and multi-factor authentication. This design presents a multi-step system combining facial recognition and OTP verification. Choosers register with particular details and a facial image. At voting time, the system uses a webcam with a Haar Cascade Classifier and LBPH algorithm to corroborate the stoner's identity. A valid match triggers an OTP transferred to the registered contact. After OTP evidence, the namer can pierce the dashboard and submit their vote, which is securely timestamped.

The SSRN paper[3] supports enhanced facial recognition delicacy using deep literacy, while the IJCRT paper[4] promotes a contactless voting approach. Blockchain integration further strengthens translucency and data integrity. The Meta- analysis of Blockchain- Powered Electronic Voting Systems[8] outlines crucial relinquishment challenges security, translucency, and sequestration — addressed through blockchain's decentralization and invariability. The Development of a Voting Technology paper[9] proposes a modular, PBFT- grounded blockchain frame for scalable choices. SHARVOT[10] introduces secret sharing and ring autographs to cover obscurity and insure empirical issues. The Blockchain Technology- Grounded E-Voting System[7] addresses EVM excrescencies and reinforces ballot security, while the SSRN paper Online Voting System Using Blockchain[11] shows how facial recognition and OTP can integrate with blockchain for a secure, tamper- evidence platform.

II. RELATED WORKS

Several exploration sweats have explored colorful executions of secure voting systems using biometric and digital technologies. The paper named " A Secure Online Voting System Using Face Recognition Technology"[1] presents a voting model where face recognition is used to corroborate choosers, demonstrating how facial biometrics can reduce namer impersonation. Their approach enhances security by combining biometric confirmation with namer enrollment databases.

The work " Smart Voting sysem using face recognition and OTP" from IRJET[2] proposes an intertwined system combining facial recognition and OTP verification to increase voting delicacy and insure namer authenticity. The system utilizes the Haar waterfall algorithm for face discovery and original double pattern Histogram for recognition, furnishing a dependable means of verifying namer identity before granting access to estate a vote.

In the SSRN paper[2], the authors emphasize the use of deep literacy algorithms to strengthen face recognition delicacy in unbridled surroundings. Their perpetration discusses prostrating challenges similar as variations in lighting, facial expressions, and angles, which are pivotal for real- world deployment.

Another donation is from the IJISRT paper" *Smart Voting System Using Face Recognition and OTP Verification*"[4], which stresses the significance of a stoner-friendly interface and concentrated authentication mechanisms. This system incorporates both biometric and OTP verification ways, icing that only licit druggies can pierce the voting gate, while also minimizing the eventuality for data tampering.

The paper" *Smart Voting System using Facial Detection*"[5] farther discusses the armature of such a system, proposing a streamlined inflow for stoner confirmation and vote recording. They argue for minimum mortal intervention, enhancing system effectiveness and integrity. Their methodology includes real- time image capturing and matching with pre-registered namer images, addressing a major challenge in icing valid participation. These[6] being systems inclusively support the growing agreement that secure, digital voting platforms can transfigure the popular process by adding availability, translucency, and responsibility.

Choices are vital to republic, taking secure, effective, and transparent processes[12]. Traditional paper- grounded voting is slow, expensive, and prone to crimes and fraud. To overcome these issues, this design introduces an online voting system using facial recognition and OTP authentication, enabling remote and secure voting.[13] The system captures the namer's face via webcam and matches it with stored data. Upon successful verification, an OTP is transferred to the registered mobile or dispatch. Only after entering the correct OTP can the namer access the dashboard and cast their vote, icing one person, one vote.

III. TECHNOLOGIES USED

A. User Authentication Module

The system includes a robust stoner authentication module that verifies the identity of both directors and choosers. Through session-grounded login operation, it ensures that only registered druggies can pierce their separate dashboards. This module plays a pivotal part in system security by precluding unauthorized access and conserving the integrity of the election process.

B. Admin Dashboard

The Admin Dashboard is a comprehensive backend panel that allows directors to manage all core functions of the voting system. From this dashboard, the admin can produce and configure new choices, define voting positions (similar as President or Secretary), register campaigners with their particular details and prints, and enroll choosers into the system. The admin also has access to view and manage results once voting is complete. The interface is designed for ease of use, offering real- time updates and announcements, icing that managing a digital election is both effective and stoner-friendly.

C. Voter Dashboard

Once authenticated, choosers gain access to a substantiated dashboard where they can view forthcoming and ongoing choices. This module is streamlined to guide druggies through the voting process easily and securely. Choosers can see the list of campaigners running for different positions and are allowed to cast their votes only formerly per election. This interface simplifies namer commerce and ensures a smooth stoner experience. The system ensures each vote is tied to a unique namer ID, precluding multiple voting and maintaining voting integrity.

D. Voting and Vote Casting Module

This module forms the heart of the operation. It presents a dynamic voting panel grounded on the election presently in progress. Choosers are shown the campaigners for each position and are allowed to elect only one. Once a vote is submitted, the system performs a check to confirm that the namer hasn't formerly suggested in that election. However, the vote is securely stored in the database, If valid. Choosers admit a evidence communication that their vote has been successfully recorded. No further variations are allowed once the vote is cast, icing non-repudiation and trust in the process.

E. Election & Candidate Management Module

This module allows the director to handle all tasks related to election setup and candidate configuration. choices can be created with custom titles and time frames. contenders can be added with necessary details, assigned to specific positions, and made visible to pickers during the election period. This module ensures that all election content is manageable in real- time and that any variations reflect roundly across the system. It provides strictness in managing multiple choices and multiple places within a single frame.

F. Result Calculation and Reporting Module

After advancing ends, the admin can pierce the results of each election through this module. It counts and displays the number of votes each candidate entered and presents the winner for each position. This module can be extended to include import options or graphical visualizations like bar or pie charts for enhanced clarity. It contributes to the translucence of the process by allowing real-time access to issues and enabling third- party review if necessary.

G. Database Management Module

The backbone of the system is a well- organized MySQL database, which stores structured information for druggies, choices, campaigners, and votes. It maintains referential integrity between tables to help duplication or loss of data. Sensitive data, similar as login credentials, are stored securely(with eventuality for unborn encryption). The database is optimized for effective queries, which helps reduce response time during vote casting, affect generation, and system monitoring.

H. Code Architecture and Reusability

The design follows a modular coding approach using HTML, allowing for clean separation of sense, design, and data handling. It includes applicable law blocks similar as a central database connection train, session tutor, and template heads footers. This design gospel reduces redundancy, simplifies debugging, and makes unborn advancements easier to apply. The modular codebase supports scalability and encourages stylish practices in software development.

I. Future Integration Scope

Though the current interpretation is a introductory secure online voting platform, it's designed with unborn upgrades in mind. Planned advancements include biometric- grounded face recognition using Python and OpenCV, OTP verification via SMS/ dispatch APIs, and conceivably the use of blockchain to insure inflexible vote logging. These technologies can help alleviate fraud, ameliorate availability for remote choosers, and make farther trust in digital voting systems.

IV. EXISTING SYSTEM

The subsisting online voting system represents a contemporary result that replaces numerous of the limitations associated with the traditional homemade voting process. This system allows registered choosers to cast their ballots digitally from any position using internet- enabled bias similar as computers or smartphones. As a result, it improves availability and convenience for a wide range of druggies, including individualities in remote areas, those with mobility challenges, and choosers who are unfit to reach polling stations due to health, trip, or scheduling conflicts. By barring the need for physical presence at polling cells, the system enhances overall namer participation and inclusiveness.

Security in the being system is maintained through a combination of One- Time word(OTP) authentication and facial recognition technology. Choosers are first vindicated through OTP transferred to their registered mobile number or dispatch, followed by facial authentication to insure identity delicacy.

This binary authentication system significantly reduces the threat of fraud, identity theft, and indistinguishable voting. also, the system automates vote counting in real- time, leading to hastily and more accurate election results while minimizing mortal crimes. By doing down with paper ballots and homemade counting, it not only speeds up the process but also reduces the overall cost of conducting choices.

V. SYSTEM DESIGN AND SPECIFICATIONS

The design and specifications of the proposed online advancing system serve as the foundational design for developing a secure, effective, and stoner-friendly digital election platform. This section outlines the structural and functional aspects that drive the system, including its overall armature, crucial factors, and specialized conditions. By integrating biometric face recognition and OTP- grounded verification, the system ensures accurate namer authentication while maintaining availability and translucency. The ideal is to give a flawless voting experience that minimizes the threat of fraud and enhances popular participation through technology- driven results.

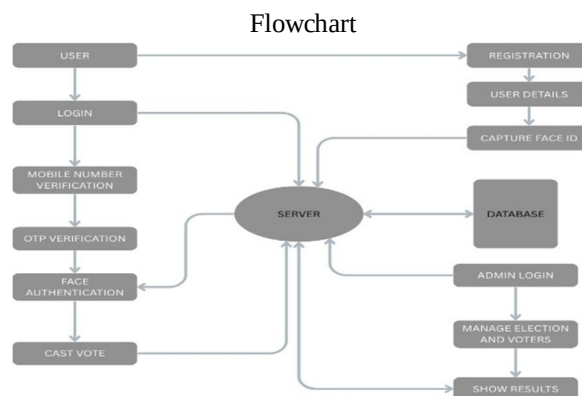


Figure 1

A. System Overview

The proposed Online Voting System aims to produce a secure and stoner-friendly platform for conducting choices digitally. By the figure 1, integrates face recognition technology and OTP- grounded authentication to corroborate the identity of choosers before allowing them to cast their votes. It targets the improvement of electoral translucency, effectiveness, and inclusiveness by enabling remote and secure participation. The system is especially salutary for individualities who are unfit to reach their native polling stations.

B. System Architecture

The architecture of the system is structured into three main components to ensure efficient functionality and security throughout the voting process. In figure 1, the first component is the User Authentication Module, which handles user login and registration, submission of facial ID, mobile number verification, OTP validation, and facial recognition using real-time webcam input. This module ensures that only verified users gain access to the voting system. The second component is the Voting Module, where the system verifies user details, presents the list of candidates, allows the voter to select and confirm their preferred candidate, and finally cast the vote securely. The third component is the Admin and Management Module, which provides administrative control over managing elections, voter and candidate records, monitoring system activity, and handling the result calculation and publication process. Together, these modules work seamlessly to deliver a secure, transparent, and user- friendly online voting system.

C. Functional Specifications

The system's functional specifications are designed to insure secure, stoner-friendly, and effective online voting operations. The process begins with namer enrollment, where druggies give particular information and upload a facial image for unborn authentication. For login, the system uses real- time face recognition, matching the live facial input with the stored data to corroborate the namer's identity. Once authenticated, an OTP verification step is initiated, where a One- Time word is transferred to the namer's registered dispatch, adding an fresh subcaste of security before allowing access to the voting interface. In the voting module, vindicated druggies can view the list of campaigners, make a selection, and securely submit their vote. An admin dashboard is handed to election directors, enabling them to add or manage seeker biographies, examiner voting exertion, and view real- time statistics. Eventually, the system includes a result protestation point that automatically calculates and displays election results once the voting period concludes, icing speed, translucency, and delicacy in the outgrowth.

VI. METHODOLOGY

The system employs a comprehensive, multi-layered methodology that integrates biometric verification with digital authentication ways to insure a secure, dependable, and transparent electoral process. originally, choosers must register by submitting particular details similar as full name, age, public identification number, a vindicated mobile number, and a clear facial image. This information is securely stored in a MySQL database using a PHP or Python- grounded backend, which manages database operations, session running, and secure API communication. During the voting phase, the system captures a real- time image via the namer's webcam and performs facial recognition using OpenCV's Haar Cascade Classifier for face discovery and the Original Binary Pattern Histogram (LBPH) algorithm for recognition, comparing the live image to the one stored during enrollment to confirm identity.

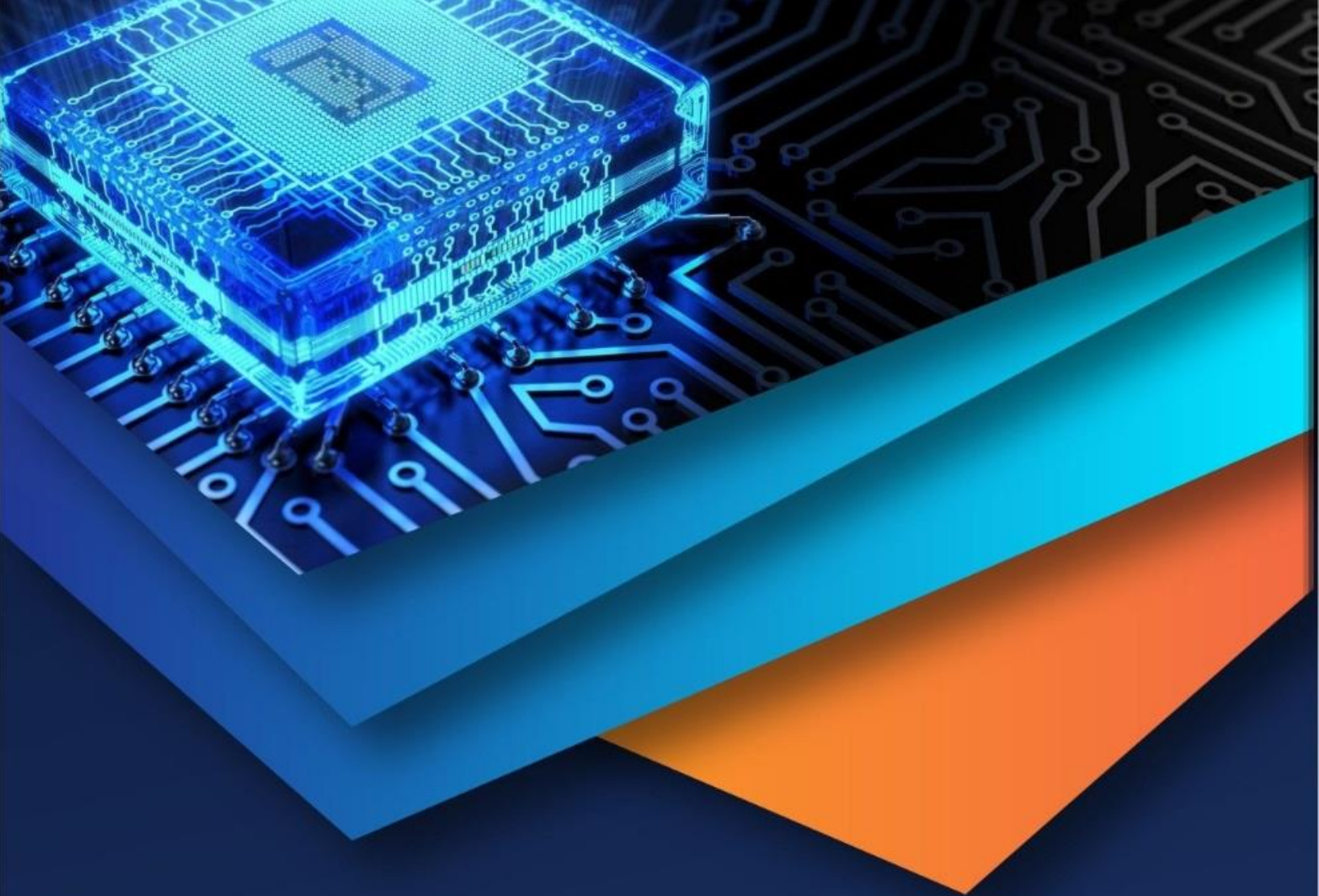
Upon successful facial authentication, the system generates a One- Time word(OTP), which is transferred to the namer's registered mobile number via an SMS gateway similar as Twilio or Fast2SMS. The namer must enter the correct OTP within a limited time window to do, therefore adding an fresh subcaste of security through two- factor authentication. Once both authentication way are completed successfully, the namer earnings access to the digital voting panel, where they can view the list of campaigners and cast their vote. The vote is also securely translated and recorded in the database, and the system ensures that each namer can bounce only formerly by streamlining a status flag linked to their profile. The platform includes a devoted director panel that allows authorized officers to cover the voting process in real time, manage access warrants, induce logical reports, and review system exertion logs, icing translucency and integrity throughout the election. Security protocols similar as HTTPS and encryption algorithms like AES or SHA are enforced to cover data during transmission and storehouse, while detailed inspection trails and logging mechanisms supportpost-election confirmation and responsibility. This integrated approach ensures a robust, stoner-friendly, and secure system for conducting online choices in a ultramodern, digital terrain.

VII. CONCLUSION AND FUTURE WORKS

The design effectively demonstrates a secure and effective online voting system that uses facial recognition and OTP verification to authenticate choosers and maintain election integrity. By barring the need for physical polling stations and homemade verification, the system reduces functional costs, minimizes mortal error, and improves availability for individualities in remote areas or with mobility challenges. The integration of biometric and digital authentication ensures a stoner-friendly and transparent voting experience. This design provides a promising step toward contemporizing the electoral process and enhancing public trust in the system. unborn advancements may include blockchain for tamper- evidence vote storehouse, advanced biometric styles like point or iris recognition, AI- grounded anomaly discovery, and bettered availability through mobile apps, multilingual support, and voice-guided interfaces. These upgrades can further strengthen security, increase namer participation, and contribute to the elaboration of a digital republic.

REFERENCES

- [1] Citra Devi Nair, Nazirah Hamid and Abdul Kadir "Describes the use of facial recognition for secure voter verification." In Proc. Of Malaysian journal of computing and applied mathematics, Terenganu, Malaysia.
- [2] Rugved Deshpande, Rutuja Boke and Dr.Rajesh Kedarnath Navandar, "ighlights the integration of OTP and facial recognition using Haar Cascade and LBPH."In Proc. Of International Researh journal of Engineering and Technology (IRJET)Mar 2024
- [3] Uma Hombal, Sanika Utpat and Shruti Yadav "Emphasizes deep learning models for face detection in varied conditions."In proceedings of International Conference on Communication and Information Processing (ICCIP-2023)
- [4] A. Manoj Kumar, K.Mithil Ready and Ass. Prof. N.Swopna Goud " Discusses multi-layered authentication for added security." In Proc. Of International Journal of Innovative Science and Research Technology (ISSN No:-2456-2165)
- [5] Chandra Keerthi Pothin , Atla Indu Reddy, Ravikumar CV, "Focuses on system architecture and real-time verification.",In Proc. Of International Journal of Innovative Technology and Exploring Engineering (IJTEE) ISSN: 2278-3075 (Online), Volume-9 Issue-6, April 2020
- [6] Ganesh Prabhu S, Prabu.S and R.R.Thirrunavukkarasu, "This system also enables the user the citizens to see the results anytime which can avoid situations that pave way to vote tampering.", International Conference on Advanced Computing and Communication Systems (ICACCS)
- [7] N. Savorona, O. Suprun, O. Provotar, T. Savorona, O. Suprun, and V. Nechyporuk, "Development of a decentralized voting system based on blockchain technology," in CH & CMIGIN 24: Third International Conference on Cyber Hygiene & Conflict Management in Global Information Networks, Kyiv, Ukraine, Jan. 2024, pp. 125–128.
- [8] V. K. Manda and M. Bhukya, "Meta-analysis of blockchain- powered electronic voting systems," MATEC Web of Conferences, vol. 392, p. 01076, 2024.
- [9] N. Navaneeth, P. Prarthana, R. G. Athreyas, and R. V. Ranjith, "Online Voting System using Blockchain," 2024.
- [10] S. Bartolucci, P. Bernat, and D. Joseph, "SHARVOT: secret SHARE-based VOTing on the blockchain," 2018.
- [11] Prof. A. A. Lahane, J. Patel, T. Pathan, and P. Potdar, "Blockchain technology based e-voting system," ITM Web of Conferences, vol. 32, 2020.
- [12] Neha Roy, Asst. Prof. Dewendra Bharambe, Nouroza Bagwan, Suryakant Godase, "using OTP generation to prevent fraud and enable location-free voting" International Journal of Advance Research and Innovative Ideas in Education (IJARIIE), Vol-9, Issue-3, 2023.
- [13] S Anbumani MCA, Mr. J. Jayapandian M.C.A, , M.Phil," remote voting using facial recognition to ensure voter authenticity," International Journal of Advanced Research in Computer and Communication Engineering.



10.22214/IJRASET



45.98



IMPACT FACTOR:
7.129



IMPACT FACTOR:
7.429



INTERNATIONAL JOURNAL FOR RESEARCH

IN APPLIED SCIENCE & ENGINEERING TECHNOLOGY

Call : 08813907089  (24*7 Support on Whatsapp)